Roll No....................

MID SEMESTER EXAMINATION 2025

Name of the Program: B. Tech
Name of the Course: Material science

Semester: 3rd
Course Code: TME 302

Time: 90 Minute

Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing any of the sub questions.
(ii) Diagrams/figures should be drawn wherever necessary.
(iii) Each question carries 10 marks.

Q.1	(10 marks)	
(a)	Differentiate between metals, ceramics, and polymers in terms of their properties. Provide suitable examples of each material type along with their common applications.	**CO1**
	OR	
(b)	Explain the different types of bonds in materials.	**CO1**
Q.2	(10 marks)	
(a)	What is a unit cell? Explain the structures of NaCl, Zinc Blende, and Fullerene with suitable diagrams	**CO1**
	OR	
(b)	How are Miller indices used to represent directions and planes in crystals? Calculate the Miller indices for a plane intersecting the x, y, and z axes at $(1, 1/2, \infty)$	**CO2**
Q.3	(10 marks)	
(a)	Explain the Bragg's law and derive the relation to determine the structure using Bragg's law.	**CO2**
	OR	
(b)	A diffraction pattern of a Body centered cubic crystal of lattice parameter a = 3.45 Å is obtained with a monochromatic x-ray beam of wavelength 1.50 Å. The first three lines on this pattern were observed to have θ values (in degrees) 23.5, 33.5, and 42.7 respectively. Determine the interplanar spacing and the miller indices of the reflecting planes.	**CO2**
Q.4	(10 marks)	
(a)	Define point defects in solids.	**CO2**
	OR	
(b)	Derive the Ficks 2nd law second form for solid state diffusion.	**CO2**
Q.5	(10 marks)	
(a)	Discuss the significance of the Hume-Rothery rules in the formation of solid solutions	**CO3**
	OR	
(b)	Discuss edge dislocation, screw dislocation, twin boundary, and volumetric defects in crystals with suitable diagrams.	**CO2**